

Student Score Analysis Using Linear Algebra

1. Introduction

In data science and machine learning, **vectors and matrices** are used to represent and analyze data.

In this project, we analyze **student subject scores** using linear algebra techniques such as:

- Vector operations
- Matrix operations
- Eigenvalues and eigenvectors
- Matrix decompositions
- Dimensionality reduction (PCA & LDA)

These techniques help us understand **relationships, patterns, and variations** in the dataset.

Objectives

1. Represent student score data using **vectors and matrices**.
2. Perform **vector operations** such as dot product, cross product, norms, and projections.
3. Apply **matrix operations** such as transpose, determinant, and inverse.
4. Understand **linear transformations and geometric interpretation**.
5. Perform **matrix decompositions (LU, SVD)**.
6. Use **PCA and LDA** to reduce dimensions and classify data.
7. Interpret **eigenvalues and eigenvectors** to understand variance in data.

Dataset Example

Assume we have **student scores in 3 subjects**:

Student	Math	Science	English
S1	85	90	78
S2	80	88	92
S3	75	85	89
S4	90	92	85
S5	70	80	75

Part A: Vector & Matrix Fundamentals

1. Represent Student Scores as Vectors

Each student's scores can be written as a **vector**.

Example:

S1 vector

$$S1 = [85, 90, 78]$$

S2 vector

$$S2 = [80, 88, 92]$$

This means each student is a **point in 3-dimensional space**.

2. Vector Norms

Norm tells the **length (magnitude)** of a vector.

Norm-1 (Manhattan Distance)

Formula

$$\|v\|_1 = |x_1| + |x_2| + |x_3|$$

Example for S1

$$\begin{aligned}\|S1\|_1 &= |85| + |90| + |78| \\ &= 253\end{aligned}$$

Norm-2 (Euclidean Distance)

Formula

$$\|v\|_2 = \sqrt{x_1^2 + x_2^2 + x_3^2}$$

Example

$$\begin{aligned}\|S1\|_2 &= \sqrt{85^2 + 90^2 + 78^2} \\ &= \sqrt{7225 + 8100 + 6084} \\ &= \sqrt{21409} \\ &\approx 146.33\end{aligned}$$

3. Dot Product

Dot product measures **similarity between two vectors**.

Formula

$$A \cdot B = x_1y_1 + x_2y_2 + x_3y_3$$

Example (S1 and S2)

$$\begin{aligned}S1 &= [85, 90, 78] \\ S2 &= [80, 88, 92] \\ S1 \cdot S2 &= 85 \times 80 + 90 \times 88 + 78 \times 92 \\ &= 6800 + 7920 + 7176 \\ &= 21896\end{aligned}$$

Higher value → students have **similar performance patterns**.

Angle Between Vectors

Formula

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

If angle is **small**, students have similar score patterns.

4. Cross Product

Cross product is used only in **3D vectors**.

Formula

$$A \times B = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Example

$$S1 \times S2 = \begin{vmatrix} i & j & k \\ 85 & 90 & 78 \\ 80 & 88 & 92 \end{vmatrix}$$

Result

$$\begin{aligned} & (90 \times 92 - 78 \times 88)i - (85 \times 92 - 78 \times 80)j + (85 \times 88 - 90 \times 80)k \\ & = (8280 - 6864)i - (7820 - 6240)j + (7480 - 7200)k \\ & = 1416i - 1580j + 280k \end{aligned}$$

5. Vector Projection

Projection shows **how much one vector lies in the direction of another**.

Formula

$$proj_B(A) = \frac{A \cdot B}{\|B\|^2} B$$

Example

$$proj_{S2}(S1) = \frac{21896}{\|S2\|^2} S2$$

This shows **how much S1 performance aligns with S2**.

Part B: Matrix Operations

4. Student Score Matrix

$$X = \begin{bmatrix} 85 & 90 & 78 \\ 80 & 88 & 92 \\ 75 & 85 & 89 \\ 90 & 92 & 85 \\ 70 & 80 & 75 \end{bmatrix}$$

Rows = Students
Columns = Subjects

Matrix Addition

Example

$$A + B = [a_{ij} + b_{ij}]$$

Each element is added.

Matrix Multiplication

Rule

$$(m \times n) \times (n \times p)$$

Example

$$X^T X$$

Used in **covariance calculations**.

Transpose

Transpose flips rows and columns.

$$X^T = \begin{bmatrix} 85 & 80 & 75 & 90 & 70 \\ 90 & 88 & 85 & 92 & 80 \\ 78 & 92 & 89 & 85 & 75 \end{bmatrix}$$

Determinant

Determinant shows **whether matrix is invertible**.

Example

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

If determinant = **0**, inverse does not exist.

Matrix Inverse

Inverse is used to **solve linear equations**.

Condition

$$\det(A) \neq 0$$

Formula

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

Part C: Linear Transformations & Geometry

Line

1D data representation.

Example

Single subject score.

Plane

2D representation.

Example

Math vs Science.

Each student becomes a **point on a plane**.

Hyperplane

In higher dimensions (3 or more).

Example

Math + Science + English → **3D space**

In machine learning, hyperplanes are used for **classification boundaries**.

Part D: Eigenvalues & Decomposition

7. Covariance Matrix

Covariance measures how variables change together.

Example covariance matrix

$$C = \begin{bmatrix} 62.5 & 48.2 & 40.3 \\ 48.2 & 45.1 & 30.7 \\ 40.3 & 30.7 & 50.6 \end{bmatrix}$$

Eigenvalues and Eigenvectors

Formula

$$Av = \lambda v$$

Where

A = matrix
v = eigenvector
 λ = eigenvalue

Eigenvalues show **variance amount in each direction**.

Largest eigenvalue → **most important feature direction**.

8. LU Decomposition

LU decomposition breaks matrix into

$$A = LU$$

Where

L = Lower triangular matrix

U = Upper triangular matrix

Example

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix}$$
$$U = \begin{bmatrix} 4 & 3 & 2 \\ 0 & 5 & 1 \\ 0 & 0 & 6 \end{bmatrix}$$

Used for **efficient solving of equations**.

9. Singular Value Decomposition (SVD)

SVD factorizes matrix into

$$A = U\Sigma V^T$$

Where

U = left singular vectors

Σ = singular values

V = right singular vectors

Purpose

- Noise reduction
 - Data compression
 - Dimensionality reduction
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Part E: Dimensionality Reduction

10. Principal Component Analysis (PCA)

PCA reduces dimensions while keeping **maximum variance**.

Steps

1. Standardize data
2. Compute covariance matrix
3. Find eigenvalues & eigenvectors
4. Select top eigenvectors
5. Transform data

Example

Original data

3 subjects

After PCA

2 principal components

Now dataset becomes

Student	PC1	PC2
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S1	1.2	-0.8
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S2	0.9	0.4
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S3	-0.3	1.1
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Meaning

Data simplified while preserving information.

11. Linear Discriminant Analysis (LDA)

LDA is used for **classification**.

Goal

Separate groups with **maximum distance between classes**.

Example classification

Student Average		Category
S1	84	Above Average
S2	86	Above Average
S3	83	Below Average
S4	89	Above Average
S5	75	Below Average

LDA finds a **projection line** that separates these groups.